



UNIVERSITAT POLITÈCNICA DE CATALUNYA
BARCELONATECH

Escola Tècnica Superior d'Enginyeria
de Telecomunicació de Barcelona



Thesis title

Master Thesis
submitted to the Faculty of the
Escola Tècnica d'Enginyeria de Telecomunicació de Barcelona
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by
Student Name

In partial fulfillment
of the requirements for the master in
(*Write the name of your Master*) **ENGINEERING**

Advisor: name of the advisor
Barcelona, Date XXXXX



Contents

List of Figures	3
List of Tables	3
1 Introduction	6
1.1 Gantt Diagram	6
1.2 Topic	6
2 State of the art of the technology used or applied in this thesis:	7
2.1 Topic	7
2.2 Topic	7
3 Methodology / project development:	8
4 Results	9
5 Budget	11
6 Environment Impact (Optional)	12
7 Conclusions and future development:	13
References	14
Appendices	15

List of Figures

1	Project's Gantt diagram	6
2	Prototype setup	9

Listings

List of Tables

1	This is the caption	9
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Revision history and approval record

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0	dd/mm/yyyy	Document creation
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DOCUMENT DISTRIBUTION LIST

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Position	Project Author	Position	Project Supervisor

Abstract

Every copy of the thesis must have an abstract. An abstract must provide a concise summary of the thesis. In style, the abstract should be a miniature version of the thesis: short introduction, a summary of the results, conclusions or main arguments presented in the thesis. The abstract may not exceed 150 words for a Degree's thesis.

1 Introduction

An Introduction that clearly states the rationale of the thesis that includes:

1. Statement of purpose (objectives).
2. Requirements and specifications.
3. Methods and procedures, citing if this work is a continuation of another project or it uses applications, algorithms, software or hardware previously developed by other authors.
4. Work plan with tasks, milestones and a Gantt diagram.
5. Description of the deviations from the initial plan and incidences that may have occurred.

The minimum chapters that this thesis document should have are described below, nevertheless they can have different names and more chapters can be added.

1.1 Gantt Diagram

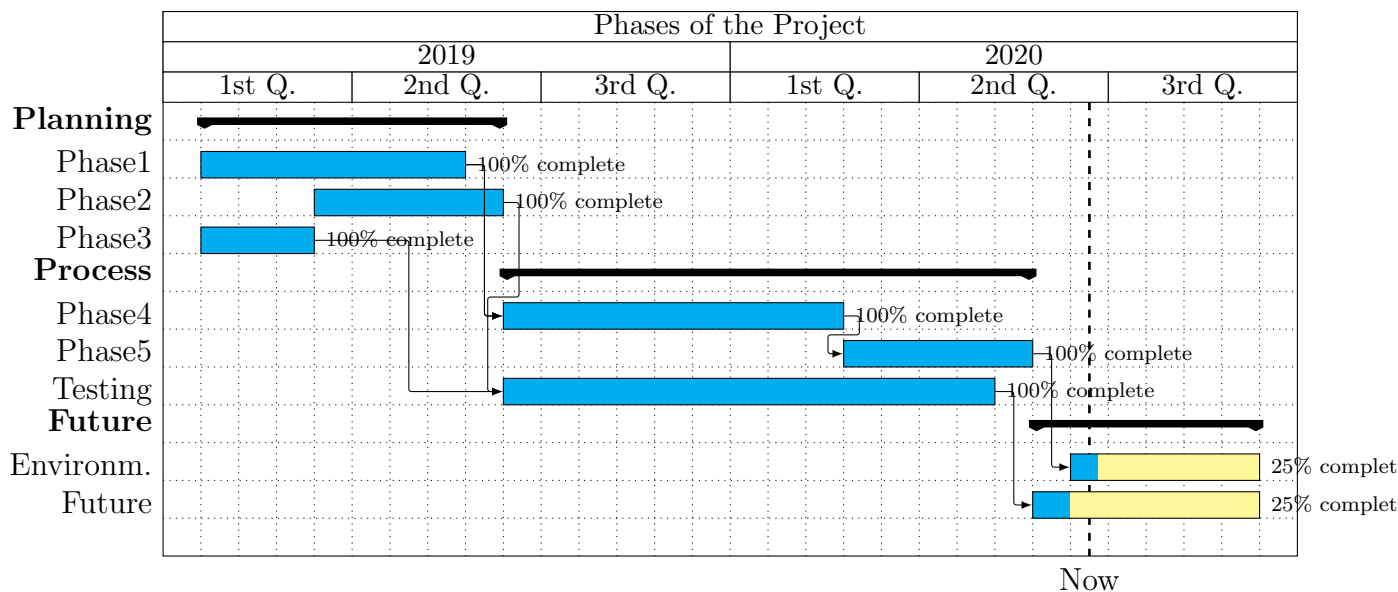


Figure 1: Gantt diagram of the project

For more information read the manual [1] of Skala.

1.2 Topic

2 State of the art of the technology used or applied in this thesis:

A background, comprehensive review of the literature is required. This is known as the Review of Literature and should include relevant, recent research that has been done on the subject matter.

2.1 Topic

Here you have a couple of references about LaTeX [3] and electrodynamics [2].

2.2 Topic

3 Methodology / project development:

The Methodology is included in this chapter and should include all relevant methods that were utilized as well as research methods and measurements, software and hardware development, ...

4 Results

This should include your data analysis and findings

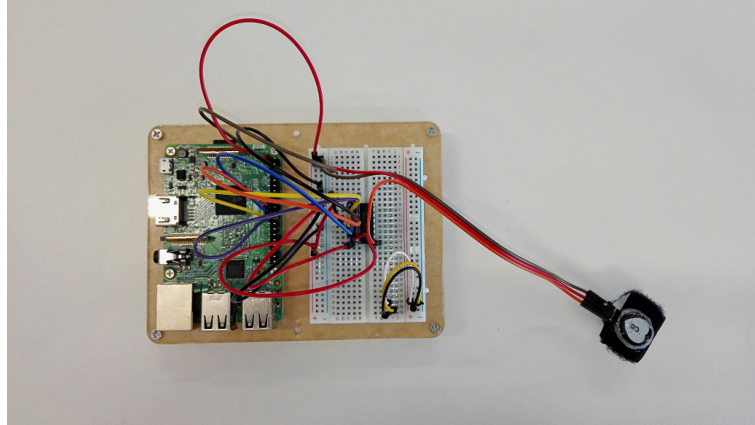


Figure 2: Prototype setup.

Table 1: This is the other caption. Since the trial size of the experiments showed is one second, the number of *Target* and *Impostor* data corresponds to number of trials or seconds

Dataset	Label	Train	Validation	Develop	Test
First	Target	135	45	30	30
	Impostor	5,220	1,740	1,890	2,880
	#Subjects	31			12
Second	Target	144	80	48	48
	Impostor	2,014	1,119	1,343	1,545
	#Subjects	15			5

Algorithm 1 Temperature-Distributed algorithm

```

1: procedure TEMP-SPREAD( $GN_i, HN_j, temperatures$ )  $\triangleright$  Lowest temperature priority
2:    $temperature\_list \leftarrow short(temperatures)$ 
3:    $max\_temperature \leftarrow max(temperature\_list)$ 
4:    $ThresHold \leftarrow 0.5$ 
5:    $temperature\_impact \leftarrow 0.2$ 
6:   for  $GN_i$  in  $i = 1, 8$  do  $\triangleright$  Iterate every hardware node on the given GN
7:      $it\_temperature \leftarrow temperature\_list(GN_i)$ 
8:      $temp\_weight \leftarrow \frac{max\_temperature - it\_temperature}{max\_temperature} * temperature\_impact$ 
9:      $\omega(Master - GN_i) \leftarrow ThresHold * temp\_weight$ 
10:    for  $HN_j$  in  $j = 1, n$  do
11:      if  $available\_accel_{i,j} > busy\_accel_{i,j}$  then
12:         $policy_\omega = \frac{AvailableHW}{TotalHW} * ThresHold$ 
13:         $\omega(GN_i - HN_{i,j}) \leftarrow ThresHold + policy_\omega$ 
14:      else
15:         $\omega(GN_i - HN_{i,j}) \leftarrow 1$ 
16:     $node \leftarrow find\_djistra\_shortest\_path(Master\_Node, aux\_node)$ 
17:    return  $node$   $b$   $\triangleright$  The gcd is b

```

5 Budget

Depending on the thesis scope this document should include:

6 Environment Impact (Optional)

Whether the tasks that have led to the realization of this thesis, as if its results have identifiable environmental impact, describe it in this section.

7 Conclusions and future development:

This should include your summary, conclusions and recommendations.

References

- [1] Wolfgang Skala. Drawing gantt charts in latex with tikz.
- [2] Albert Einstein. Zur Elektrodynamik bewegter Körper. (German) [On the electrodynamics of moving bodies]. *Annalen der Physik*, 322(10):891–921, 1905.
- [3] Michel Goossens, Frank Mittelbach, and Alexander Samarin. *The L^AT_EX Companion*. Addison-Wesley, Reading, Massachusetts, 1993.
- [4] Donald Knuth. Knuth: Computers and typesetting.

Appendices

Appendices may be included in your thesis but it is not a requirement.